

SVENNER FYR, Norway

WMO: 014780

Lat: 58.969N

Lon: 10.148E

Elev: 15

StdP: 101.14

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.9	-8.1	N/A	N/A	N/A	N/A	N/A	N/A	17.1	3.6	16.2	4.2	5.8	10	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.8	22.9	N/A	21.2	N/A	20.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.8	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	15.2	13.7	DB	-9.6	24.8	2.7	2.1	-11.5	26.3	-13.1	27.5	-14.6	28.7	-16.5	30.2	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	0.4	0.2	3.1	6.3	11.3	14.9	17.5	16.7	13.8	9.2	5.0	2.3		
	DBStd	6.83	3.80	3.81	2.99	2.70	3.55	2.05	2.21	1.59	2.01	2.73	3.62	4.62		
	HDD10.0	1367	299	275	214	116	24	0	0	0	1	47	151	238		
	HDD18.3	3638	557	508	472	361	221	106	43	56	134	282	400	497		
	CDD10.0	797	0	0	1	6	64	147	232	207	117	23	1	0		
	CDD18.3	26	0	0	0	0	2	2	17	4	0	0	0	0		
	CDH23.3	34	0	0	0	0	4	2	28	0	0	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0		
(14) Wind	WSAvg	6.5	7.3	7.1	6.3	5.5	5.5	5.7	5.5	6.0	6.8	7.1	7.3	7.4		
(15) Precipitation	PrecAvg	1020	93	68	58	58	64	85	84	101	96	111	106	96		
	PrecMax	1407	190	171	140	114	104	165	182	167	193	248	281	183		
	PrecMin	789	21	32	9	19	12	12	11	26	40	22	41	23		
	PrecStd	149	41	34	30	27	27	32	41	36	42	51	50	41		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	7.8	12.9	16.2	23.4	22.6	25.9	22.4	18.7	15.1	12.1	10.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	7.0	6.0	9.1	13.9	20.4	20.5	23.9	21.0	17.8	14.4	11.1	9.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	6.3	5.2	7.4	12.1	18.2	19.1	22.2	20.1	17.2	13.9	10.4	8.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	5.4	4.5	6.4	10.4	16.4	17.9	20.8	19.3	16.7	13.2	9.7	7.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		3.2	3.0	3.9	4.4	4.4	3.9	3.8	3.6	3.6	3.5	3.2	3.4		
	5% DB	MCDBR	3.9	3.6	5.7	6.9	6.8	5.5	5.8	4.6	3.8	3.2	3.4	3.2		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.288	0.323	0.341	0.380	0.373	0.359	0.381	0.378	0.349	0.337	0.304	0.264		
	taud		2.244	2.323	2.369	2.317	2.414	2.477	2.436	2.438	2.505	2.432	2.292	2.172		
	Ebn at Noon		527	696	799	823	858	877	846	820	790	676	501	415		
	Edh at Noon		44	68	86	107	104	99	102	95	76	61	42	33		
(44) All-Sky Solar Radiation	RadAvg		0.32	0.91	2.40	3.91	5.16	5.88	5.30	4.14	2.73	1.23	0.43	0.22		
	RadStd		0.06	0.18	0.23	0.54	0.53	0.39	0.59	0.28	0.31	0.20	0.10	0.03		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (53 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page